Multilevel Feature Adaptation and Aggregation of Vision Foundation Model for Remote Sensing Scene Classification

Chonghua Lv, *Student Member, IEEE*, Dan Zhang, Dou Quan, *Member, IEEE*, Ning Huyan, *Member, IEEE*, Jiaqi Chen, Shuang Wang, *Senior Member, IEEE*, and Licheng Jiao, *Life Fellow, IEEE*

***Abstract*—Remote sensing scene classification (RSC) is crucial for image content interpretation and land function identification. Vision foundation models (VFMs) have strong general visual representation capabilities, which will enhance the feature extraction from remote sensing images and improve scene classification performances. However, there are obvious discrepancies in datasets and tasks between VFMs and RSC, which will impact the accuracy of scene classification. This article proposes a VFM-based framework with a multilevel feature adaptation and aggregation approach, labeled as VFM-FAA, for RSC, which can effectively exploit and utilize VFMs' advantages for the downstream RSC task. Specifically, VFM-FAA adapts the VFM of DINOV2 for RSC from the perspectives of the dataset and task characteristics of the downstream task, respectively. First, considering that there are diverse objects with various sizes in remote sensing images, this article builds a multiscale adapter (MSA) based on the frozen VFM, thereby boosting feature extraction from remote sensing images. Second, this article proposes a multilevel feature aggregation (MFA) module to predict the scene category, which can comprehensively capture the discriminative semantic features of various scenes and further improve scene classification accuracy. Experimental results on three public datasets have demonstrated the efficiency and significant advantages of the proposed method on RSC. The proposed method can effectively enhance the generalization of the VFM in the RSC task and reduce the reliance on labeled training data, which has distinct advantages when there is a small amount of labeled training data. Compared with existing methods, VFM-FAA acquires accuracy performance gains by 2.78% in the AID dataset with 20% training samples, and improves the classification accuracy by 1.93% and 2.48% in the NWPU-RESISC45 dataset with 10% and 20% training samples, respectively.**

***Index Terms*—Multilevel feature aggregation, multiscale adapter (MSA), remote sensing images, scene classification, vision foundation models (VFMs).**

Received 9 March 2026; accepted 8 April 2026. Date of publication 13 April 2026; date of current version 23 April 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62201407, Grant 62501356, and Grant 62271377; in part by the Postdoctoral Fellowship Program of China Postdoctoral Science Foundation (CPSF) under Grant GZC20251172; in part by CPSF under Grant 2022M722496; in part by the Key Research and Development Program of Shanxi Program under Grant 2024GX-ZDCYL-02-17 and Grant 2024GX-ZDCYL-02-08; and in part by the Key Scientific Technological Innovation Research Project by the Ministry of Education. *(Corresponding authors: Dou Quan; Ning Huyan.)*

Chonghua Lv, Dan Zhang, Dou Quan, Jiaqi Chen, Shuang Wang, and Licheng Jiao are with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education of China, School of Artificial Intelligence, Xidian University, Xi'an 710071, China (e-mail: dquan@stu.xidian.edu.cn).

Ning Huyan is with the Department of Automation, Tsinghua University, Beijing 100084, China (e-mail: n-hy@mail.tsinghua.edu.cn).

Digital Object Identifier 10.1109/TGRS.2026.3683138

I. Introduction

REMOTE sensing scene classification (RSC) plays an important role in remote sensing image applications, such as land use monitoring, disaster assessment, and environmental protection [1]. RSC aims to identify the corresponding land function and predict semantic categories according to their spatial information and surface features of remote sensing images [2]. However, remote sensing images usually contain complex objects in image scenes. There are significant differences in the same scene caused by seasonal changes, illumination changes, viewpoint changes, and background changes, and high similarities across scenes brought by similar background and surface features, which increase the difficulty of scene classification.

Existing deep learning-based methods try to use a deep convolutional neural network [3], [4], or a transformer framework [5] to extract discriminative features from remote sensing images, and then use a classifier to distinguish scene semantic classes according to the extracted deep features. Due to their strong feature representation ability, deep learning methods have shown significant advantages over handcrafted features in RSC [6]. However, as shown in Fig. 1, existing methods usually train all network parameters of the deep model from scratch based on a lot of labeled training data, leading to expensive training and data labeling costs. When the number of labeled training data is limited, the performance of the deep model will significantly degrade.

Recently, vision foundation models (VFMs) that were optimized by self-supervised learning based on large-scale datasets have shown strong general feature representation and generalization, such as BYOL [7], SAM [8], and DINOv2 [9]. VFMs are also introduced into remote sensing image processing [10], [11], [12], such as image classification [13], [14], image change detection [15], semantic segmentation [16], [17], and object detection [18]. First, researchers aim to leverage the advantages of pretrained VFMs based on natural images for enhancing remote sensing image processing performances in several following perspectives: generate remote sensing image annotation (i.e., FMARS [19] and Text2Seg [17]), using general visual feature representation (i.e., FM-ReSCD [15] and RSBuilding [20]), and robust cross-modal and cross-scenes generalization (i.e., LFMDA [21]). In addition, several works construct a general remote sensing foundation model by collecting a large number of remote sensing datasets and

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:47:23 UTC from IEEE Xplore. Restrictions apply.

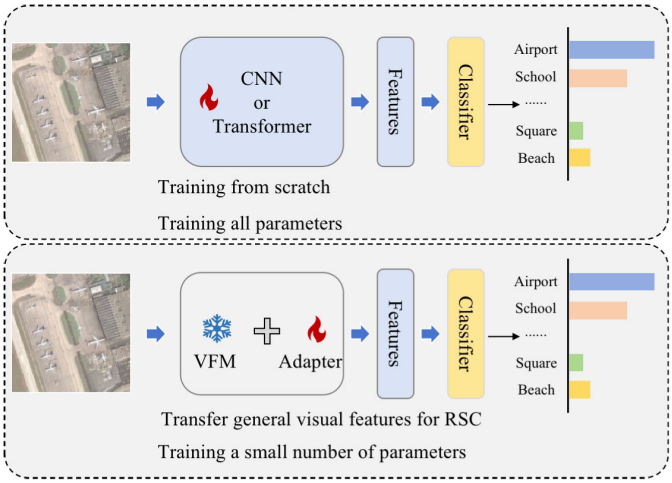


Fig. 1. Existing deep learning-based methods usually adopt a deep model to extract features for RSC and train all network parameters from scratch. This article aims to transfer the general visual features from the VFM and train a small number of parameters for adapting the RSC task.

pretraining the foundation model via self-supervised learning, such as SpectralGPT [22], SkySense [23], RingMo [24], and RingMoGPT [18].

Unlike existing deep learning-based methods that train all parameters from scratch, this article aims to utilize the general feature representation of VFM and train a small number of parameters for adapting the RSC task, thereby decreasing the reliance on a lot of labeled datasets and reducing training costs. However, there are significant differences in datasets and tasks between the pretrained VFM and RSC, limiting the performance of the VFM on RSC. To efficiently exploit the advantages of the VFM for RSC, this article proposes a multilevel feature adaptation and aggregation approach, labeled as VFM-FAA. Specifically, VFM-FAA adapts the VFM of DINOV2 for RSC from two perspectives: dataset distribution adaptation and task adaptation. First, VFM-FAA introduces a multiscale adapter (MSA) based on the frozen DINOV2, which can capture various object information with different sizes in remote sensing images and enhance the feature extraction for accurate RSC. Second, VFM-FAA proposes a multilevel feature aggregation module to predict the scene category, which fuses the comprehensive scene features and further boosts the scene semantic classification performances.

The main contributions of this article are as follows.

1) This article proposes a visual foundation model framework with multilevel feature adaptation and aggregation for RSC, labeled as VFM-FAA. It can efficiently leverage the strong general visual feature representation capability of the VFM for improving RSC performances and reducing training data labeling and optimization costs.
2) VFM-FAA addresses the significant differences in datasets and tasks between the pretrained VFM and RSC by an MSA and a multilevel feature aggregation module. VFM-FAA can capture multiscale and multilevel features of images, enhancing the scene classification accuracy.
3) Extensive experiments on various public datasets have shown the effectiveness and significant advantages of the proposed VFM-FAA over existing deep learning-based methods, especially under limited labeled training datasets. Furthermore, this article presents visualization results to demonstrate the feasibility of using VFM for RSC and the effectiveness of our proposal for adapting the VFM for scene classification.

II. Related Work

A. Remote Sensing Scene Classification

Remote sensing image interpretation usually contains pixel-level (i.e., semantic segmentation), object-level (i.e., object detection), and scene-level processing (i.e., scene classification) [1]. RSC aims to identify the scene category of images, which focuses on the global content understanding of remote sensing images [25].

Existing deep learning-based methods mainly adopt the convolutional network [3], [4], [26] or transformer structure [5], [27], [28] to extract features for scene category distinguishing, which have high intraclass invariance and interclass separability. However, there are several challenges in RSC. For example, high diversity in intraclass will increase the difficulty of scene category classification, and high similarity in interclass will bring misclassification. In addition, there are significant differences in the size of various scenes and multiple objects in complex scenes, significantly improving the challenge of scene classification [29], [30].

Existing deep learning-based methods mainly improve the scene classification performances from several aspects, including 1) extracting discriminative features through multiscale and multilevel feature fusion [31], [32], [33]; 2) leveraging the relationship of samples or scene categories for improving classification results [34]; 3) enhancing classification performances under limited labeled training samples or few-shot scene classification [35]; and 4) bringing extra knowledge for boosting scene classification accuracy [36].

For example, to deal with the large-scale variation of objects in remote sensing images, Wang et al. [29] use a global–local two-stream architecture to extract global features and local features for scene classification. Zhao et al. [37] adopt a progressive aggregation with local and global cooperative learning for effectively distinguishing scene categories. Wang et al. [38] use an enhanced feature pyramid network to extract multiscale and multilevel features and perform deep feature fusion for scene classification. MSA-Network [39] proposes a multiscale attention network, which extracts multiscale features and employs the attention learning module to select the most relevant regions for boosting remote sensing scene image classification. Chavda and Goyani [40] propose two-level feature extraction with double channel–spatial attention residual blocks for scene classification. Zhang et al. [41] propose a multiple hierarchical cross-scale transformer model to capture multiscale features and multilevel cross-scale feature representations for scene classification. SFIN [31]

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:47:23 UTC from IEEE Xplore. Restrictions apply.

proposes a semantic-assisted feature integration network for scene classification, which adopts a dual-scale feature extractor module to extract multiscale features, and uses a local semantic enhancement module and a cross-scale interactive attention module to capture the relationships across different scales. EMTCAL [5] proposes an efficient multiscale transformer and cross-level attention learning to fuse the multilevel features for improving scene classification performances. DBNet [42] builds a dual-branch network to capture multiscale and multi-angle information and deals with the scarce semantic labels through class-level prototype-guided learning. MLMamba [43] adopts a Mamba model to capture local features, multiscale features, and long-range dependencies for more accurate RSC.

Meanwhile, several works leverage the relationships between image samples or scene categories for improving RSC. SAGN [34] proposes a semantic-aware graph network to extract the multiscale information, capture diverse semantics, and exploit their relations for enhancing scene classification. HHTL [28] adopts a homo–heterogenous transformer learning framework to extract features and mine the similarities between scenes for boosting the separability of remote sensing scenes. EMSCNet [2] proposes an efficient multisample contrastive network to mine the similarity and discriminative knowledge between samples based on the positive and negative pairs.

To deal with the limited labeled training samples, SPG-GAN [44] adopts the generative adversarial network to generate labeled samples for improving the classification accuracy of remote sensing image scenes. DCN [45] uses a dual contrastive network for few-shot remote sensing image scene classification, which builds two auxiliary supervised contrastive learning branches for network optimization, including a context-guided contrastive learning branch to improve the discrimination of interclass, and a detail-guided contrastive learning branch to capture the details in intraclass samples. HiReNet [46] proposes a hierarchical-relation network for few-shot scene classification, which contains an embedding network for feature extraction and a relation network for capturing hierarchical relations among query and support samples. Li et al. [47] propose locality-preservation deep cross-modal embedding networks for zero-shot RSC. Liu et al. [48] aim to mine contrastive relations between cross-modal features and perform multilevel feature alignment for zero-shot RSC.

In addition, Chai et al. [49] integrate textual semantic information to boost scene performances through the dynamic soft label. PADA-Net [36] proposes a prompt-integrated adversarial unsupervised domain adaptation framework for scene classification, which improves the semantic association between images and text and enhances the feature discriminability.

B. VFMs for Remote Sensing Images

VFMs usually adopt a large number of datasets to train a deep model by a self-supervised learning approach, such as contrastive learning, predicting the image contents based on the masked images, predicting the transformation of images, and so on. VFMs have strong general feature representation and generalization, such as BYOL [7], SimCLR [50], SimMIM [51], SAM [8], and DINOv2 [9].

Recently, VFMs have applied for remote sensing image classification [13], [14], [22], remote sensing change detection [15], remote sensing semantic segmentation [16], [17], and remote sensing object detection [18]. VFM-based methods for remote sensing image processing mainly contain two classes: 1) using the pretrained VFM based on natural images for remote sensing, and 2) pretraining a VFM based on large-scale remote sensing datasets for remote sensing applications. On the one hand, researchers utilize the VFM to enhance image annotation and reduce the training data labeling cost. On the other hand, several works aim to leverage the general visual feature representation ability and robust generalization on cross-modal or cross-scenes of the VFM for improving the remote sensing processing. For example, FMARS [19] employs the foundation models for remote sensing image annotation. Text2Seg [17] uses different VFMs to generate prompts, reducing the need for fully annotated datasets in semantic segmentation. FM-ReSCD [15] utilizes the segment anything model (SAM) to extract features from remote sensing images and enhance the semantic change detection performances. RSPrompter [16] proposes a prompt generation method for the SAM and applies it to remote sensing image instance segmentation. LFMDA [21] leverages the powerful generalization ability of VFMs for cross-domain remote sensing semantic segmentation. LFMDA transfers the knowledge of VFMs to the student model through knowledge distillation, thereby extracting domain-invariant and discriminative features in remote sensing images and improving cross-domain remote sensing semantic segmentation accuracy. VSGNet [52] proposes a multimodal fusion framework for boosting remote sensing semantic segmentation, which uses the vision structure guidance derived by the VFM to deal with the misalignment across image modality. Cloud-adapter [53] proposes a parameter-efficient adaptive approach to adapting the pretrained VFM for enhancing the performance of cloud segmentation. RSBuilding [20] extracts features from images based on the prior knowledge of the foundation model for remote sensing image building interpretation.

In addition, several works pretrain a general foundation model for remote sensing image processing by self-supervised learning based on a large number of remote sensing datasets. For example, Wang et al. [14] build large vision transformers with about 100 million parameters for remote sensing tasks, object detection, classification, and segmentation tasks. SpectralGPT [22] proposes a universal remote sensing foundation model with a 3-D generative pretrained transformer GPT for spectral remote sensing images. The pretrained SpectralGPT achieves significant performance in spectral remote sensing image applications. SkySense [23] builds a multimodal remote sensing foundation model based on multimodal optical and synthetic aperture radar images. RingMo [24] proposes a remote sensing foundation model based on generative self-supervised learning, which can use the general visual feature representation capabilities of the foundation model for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:47:23 UTC from IEEE Xplore. Restrictions apply.

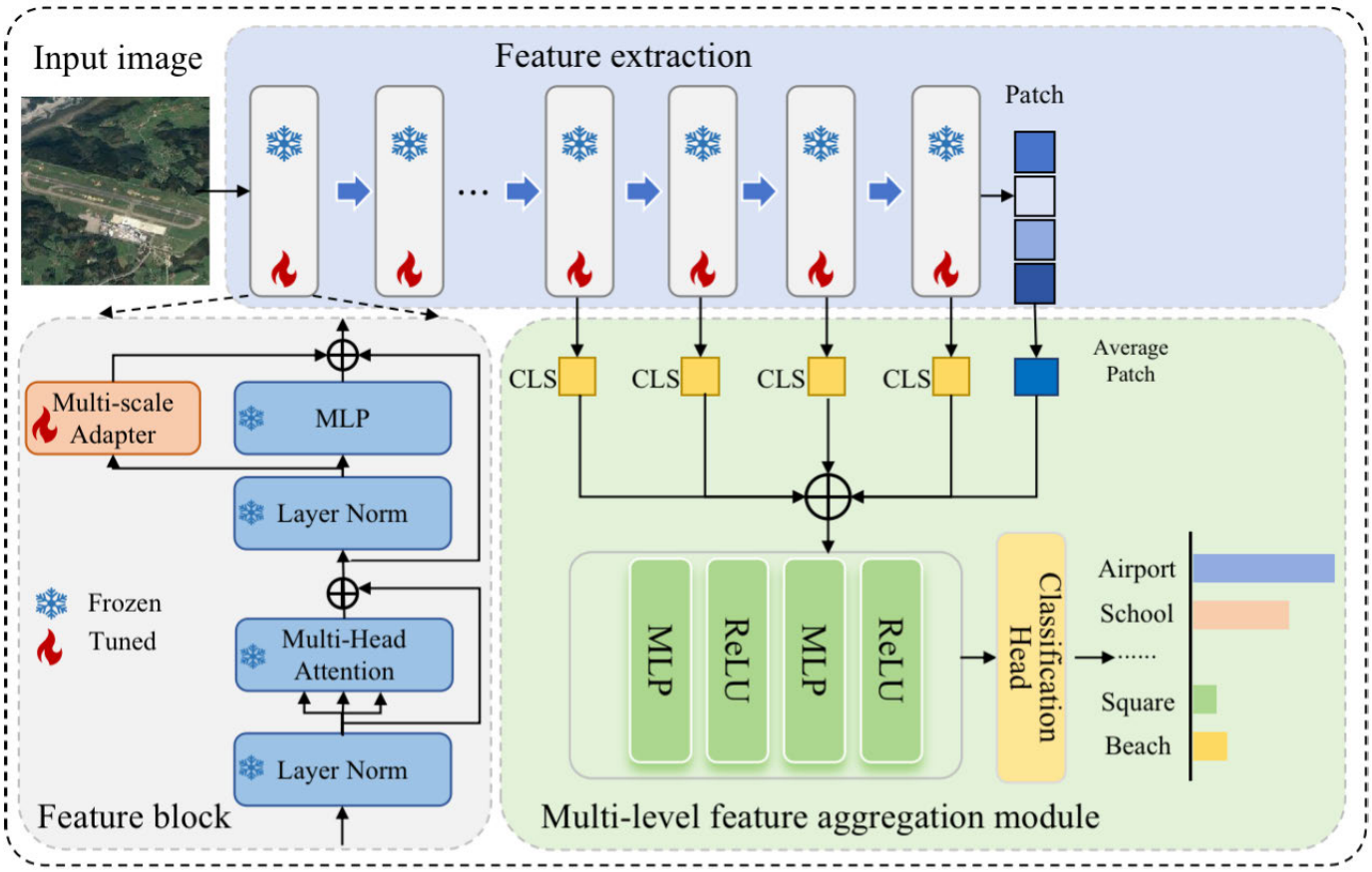


Fig. 2. Framework of the proposed VFM-based network with multilevel feature adaptation and aggregation for RSC, VFM-FAA. It transfers the general visual feature representation of VFM for RSC, performs multiscale feature adaptation, and conducts multilevel feature aggregation for more precise scene classification.

improving remote sensing image processing and applications. RingMoGPT [18] proposes a multimodal foundation model for various remote sensing tasks.

C. Discussion on Differences and Advantages

Unlike existing methods that train a randomly initialized deep network from scratch, this article aims to leverage the general visual feature representation of the VFM for enhancing the RSC. On the one hand, the VFM-based framework can reduce the training cost of the model, since we freeze the pretrained parameters of VFM and only train a small number of parameters for adapting RSC. On the other hand, our proposed deep model can reduce dependence on training data and achieve better performance with limited labeled training data. The core issue of this article is how to effectively utilize the general visual feature representations and adapt them to be more suitable for the downstream task of scene classification.

III. Methods

A. Problem Formulation

RSC aims to identify the semantic category of the input images, which usually represents their corresponding land function.

Suppose that the input images are $\{X_i\}$, we can employ a deep model to predict their scene category, $\{\hat{Y}_i\}$, which can be formulated as follows:

$$\hat{Y}_i = P(X_i), i = 1...N \tag{1}$$

where P represents the scene category prediction model, and N is the number of remote sensing images.

The deep model can be optimized using cross-entropy loss, which can be represented as follows:

$$L_{CE} = -\sum_{i=1}^{N} y_i log(\hat{y}_i) \tag{2}$$

where L_{CE} is the cross-entropy loss, and y_i is the ground-truth of scene category.

B. Overall Framework

The overall framework of the proposed VFM-based network with multilevel feature adaptation and aggregation, VFM-FAA, for RSC is illustrated in Fig. 2. It consists of four modules: a frozen pretrained VFM of DINOV2, an MSA, a multi-level feature aggregation (MFA) module, and a classification head (CH).

VFM-FAA first employs DINOV2 for feature extraction from input images, which transfers the general visual feature

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:47:23 UTC from IEEE Xplore. Restrictions apply.

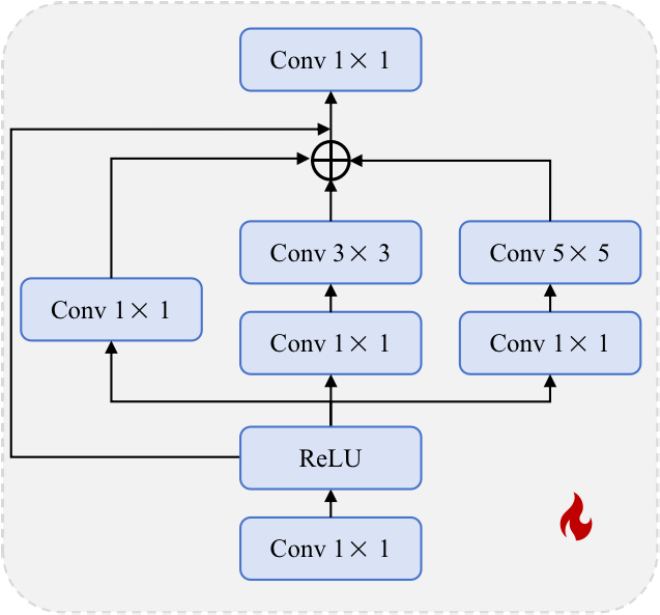


Fig. 3. Network structure of the MSA.

representation of VFM to the RSC task. We insert MSA into each Transformer block to alleviate the data distribution differences and task differences between the pretrained VFM and RSC, thereby enhancing the discriminative feature representation for remote sensing classification. Next, MFA fuses multilevel semantic information for improving classification performance. Finally, CH predicts the scene category according to the aggregated features. The details are as follows.

C. VFM for Feature Extraction

VFM-FAA aims to use the feature representation of the VFM for enhancing remote sensing scene perception. We employ the frozen pretrained DINOv2 [9] for feature extraction from remote sensing images. DINOv2 trains various visual Transformer structures by a self-supervised strategy based on the large-scale curated LVD-142M dataset, and provides general visual feature representations. Specifically, this article uses the ViT-Large/14 network structure of DINOv2 to extract multilevel features from input remote sensing images.

However, there are significant differences in data distribution and task between pretraining and RSC. If we directly applied the frozen pretrained DINOv2 for RSC, it would lead to limited scene classification accuracy.

D. Multiscale Adapter

The parameter-efficient transfer learning [54] provides an effective way to adapt VFM for downstream tasks, which freezes the pretrained model and only fine-tunes the added lightweight adapter. We insert MSA into each transformer block of VFM, adapting the feature representation of the pretrained VFM to the RSC task.

The framework of MSA is shown in Fig. 3. It employs multiple convolutional layers with varying kernel sizes for multiscale feature adaptation. MSA consists of three parallel convolutional layers with the kernel 1×1, 3×3, and 5×5. Specifically, to reduce the computation cost brought by the large convolutional kernel size, MSA adopts a convolutional layer with a kernel size 1×1 to decrease the number of channels. In addition, to prevent the degradation of feature representations in the VFM, we fuse pretrained features from the VFM and the adapted features from the MSA. The feature adaptation in each feature extraction block can be formulated as follows:

$$\begin{aligned} z_i' &= MHA\left(LN\left(z_{i-1}\right)\right) + z_{i-1} \\ z_i &= MLP\left(LN\left(z_i'\right)\right) + s \times MSA\left(LN\left(z_i'\right)\right) + z_i' \end{aligned} \tag{3}$$

where z_{i-1} and z_i are output features in the previous and current feature extraction block, respectively, MHA represents the multihead attention, LN is layer normalization operator, MLP is multilayer percetion, MSA means the multiscale adapter module, and s is a scale factor in feature fusion.

On the one hand, MSA adjusts the pretrained features to adapt to the data characteristics of remote sensing images. On the other hand, MSA can capture multiscale features and extract information from ground objects of varying sizes.

E. Multilevel Feature Aggregation

Most ViT-based classification methods typically use the CLS token for classification. However, the patch token of VFM has rich spatial information and details of images, which are often overlooked. This article adopts the MFA to capture multilevel semantic discriminative information, thereby further improving scene semantic classification performance.

CLS token contains semantic information of remote sensing scenes, while patch token has rich spatial information and details of images. MFA module fuses the CLS token in the last four layers ($f_j^{CLS} \in R^{B\times L}, j = 1,2,3,4$) and the patch token ($f^{patch} \in R^{B\times L\times D}$) in the last layer for more accurate scene category prediction. It can be represented as follows:

$$\begin{aligned} f_{avg}^{patch} &= AVG\left(f^{patch}\right) \\ f^{fused} &= FA\left(f_j^{CLS}, f_{avg}^{patch}\right) \end{aligned} \tag{4}$$

where AVG is the global average operator, and FA is the feature aggregation based on feature concatenation and multilayer perceptions, and f^{fused} is the fused feature for scene classification.

Finally, we can predict the scene category according to the fused feature through a CH

$$\hat{y}_i = CH\left(f^{fused}\right) \tag{5}$$

where CH represents the classification prediction head based on the fully connected layer.

IV. Experiments

This section verifies the effectiveness and advantages of the proposal in RSC based on three public datasets. This article first presents the advantages of the proposed method over

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:47:23 UTC from IEEE Xplore. Restrictions apply.



Fig. 4. Some remote sensing images in the UCM dataset.

existing deep learning-based methods. In addition, we test the influence of each module, such as the MSA and MFA. This part also shows the visualization results of the learned features from the remote sensing image.

A. Datasets

This article uses three public datasets for RSC evaluation, including UCM, AID, and NWPU-RESISC45 datasets.

The UCM dataset was collected by the USGS National Map Urban Area Imagery collection by the University of California, Merced. This dataset was captured from different regions, such as Boston, Columbus, Houston, Los Angeles, and Miami. It contains 2100 images from 21 semantic scene categories, where each scene category has 100 images, and each image has a size of 256 $\times$ 256 and a spatial resolution of 0.3 m. Some remote sensing scene images in the UCM dataset are shown in Fig. 4.

The AID dataset was collected by Wuhan University. It contains 10 000 remote sensing images with a size of 600 $\times$ 600 and a spatial resolution of 0.5–8 m, and has 30 semantic scene categories. This dataset has high diversity, which was captured at different times and from various countries and regions, such as China, USA, U.K., France, Italy, and so on. Some remote sensing images in the AID dataset are shown in Fig. 5.

The NWPU-RESISC45 dataset was collected by Northwestern Polytechnical University. It contains 31 500 images from 45 semantic scene categories, where each scene category has 700 images, and each image has a size of 256 $\times$ 256 and a spatial resolution of 0.2–30 m. There are various scene categories, high diversities in the same scene, and similarity across different scene categories, increasing the difficulties of the RSC. Some remote sensing scene images in the NWPU-RESISC45 dataset are shown in Fig. 6.

B. Experimental Details and Evaluation Metrics

1) Implement Details: The experiments were implemented on the Nvidia RTX4090 GPU and PyTorch framework. We adopt the Adam optimizer with a training epoch of 100, an initial learning rate of 0.00003, and a batch size of 32. The image size for feature extraction and scene classification is 224 $\times$ 224, and the scale factor $s = 0.2$ in experiments. We employ random rotation, horizontal flipping, and vertical flipping for data augmentation in experiments.

2) Evaluation Metrics: In experiments, we use the overall accuracy (OA) and confusion matrix (CM) for scene classification evaluation.

OA represents the accuracy of the scene classifier on all categories, which can be represented as follows:

$$OA = \frac{N_r}{N} \tag{6}$$

where N_r and N are the numbers of correctly classified scene images and all test images, respectively.

CM presents the probability that each class of image is predicted as all different categories, showing the accuracy of each category and the confusion between different categories.

In experiments, we report the average results and standard deviations of the five repeated experiments based on random training and test samples. This part verifies the effectiveness of the proposed method on scene classification based on different ratios of training and test datasets. Specifically, we adopt the proportions of training samples as 50% and 80% in the UCM dataset, 20% and 50% in the AID dataset, and 10% and 20% in the NWPU-RESISC45 dataset.

C. Comparison and Analysis

1) Comparison With State-of-the-Art RSC: This part compares the scene classification results of the proposal with

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:47:23 UTC from IEEE Xplore. Restrictions apply.



Fig. 5. Some remote sensing images in the AID dataset.

existing deep learning-based methods, including deep convolutional networks and transformer-based networks.

As shown in Table I, the proposed method achieves the best classification performance on the AID and NWPU-RESISC45 datasets, and obtains satisfying accuracy on the UCM dataset. In addition, our method obtains a small variance of classification accuracy and presents strong stability in RSC. For example, the OA results of the VFM-FAA in the AID dataset are 97.47% and 98.42% based on training ratios of 20% and 50%, respectively. Our proposal surpasses the HSATF by 4.19% and 0.21%, when the training ratios are 20% and 50%, respectively. In the NWPU-RESISC45 dataset, the OA results of the VFM-FAA are 94.69% and 96.15% based on training ratios of 10% and 20%, respectively, which exceed the results of HSATF by 1.93% and 2.48%. In the UCM dataset, VFM-FAA obtains a very high accuracy that is close to the best result. When the training ratios are 50% and 80%, the OA results of our VFM-FAA are 98.89% and 99.81%, respectively.

In addition, our proposal demonstrates significant advantages under a small number of training samples. For example, when the ratio of training samples is 20% in the AID dataset, the proposed method achieves an accuracy of 97.47%, significantly outperforming current state-of-the-art methods, such as HSATF (93.28%) and EMTCAL (94.69%). When the ratio of training samples is 10% in the NWPU-RESISC45 dataset, our method achieves an OA result of 94.69%, surpassing the results of HSATF by 1.93%, the accuracy result of EMTCAL by 3.06%, and the result of the GoogLeNet by 18.5%. These

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:47:23 UTC from IEEE Xplore. Restrictions apply.



Fig. 6. Some remote sensing images in the NWPU-RESISC45 dataset.

TABLE I
EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON UCM, AID, AND NWPU-RESISC45 DATASETS

Methods	Trainable Params	UCM		AID		NWPU-RESISC45	
		50%	80%	20%	50%	10%	20%
GoogLeNet [3], [6]	10.07M	92.70±0.60	94.31±0.89	83.44±0.40	86.37±0.55	76.19±0.38	78.48±0.26
VGG-VD-16 [4], [6]	134.44M	94.14±0.69	95.21±1.20	86.59±0.29	89.64±0.36	76.47±0.18	79.79±0.15
EFPN-DSE-TDFF [38]	-	96.19±0.13	99.14±0.22	94.02±0.21	94.50±0.30	-	-
MSA-Network [39]	-	97.80±0.33	98.96±0.21	93.53±0.21	96.01±0.43	90.38±0.17	93.52±0.21
ViT [55]	303.22M	97.83±0.16	98.95±0.24	93.74±0.27	96.49±0.18	90.05±0.29	92.61±0.14
T2T-ViT-12 [56]	6.64M	97.79±0.20	99.10±0.32	94.39±0.22	96.29±0.24	90.62±0.18	93.19±0.10
PVT-V2-B0 [27]	3.42M	97.94±0.44	98.86±0.38	93.52±0.35	96.27±0.14	89.72±0.16	92.95±0.09
EMTCAL [5]	27.84M	98.67±0.16	99.57±0.28	94.69±0.14	96.41±0.23	91.63±0.19	93.65±0.12
MSCT [41]	22.92M	98.97±0.11	99.84±0.16	93.28±0.26	98.21±0.39	92.76±0.13	93.67±0.29
UPetu [57]	0.65M	98.52	99.05	96.29	97.06	92.13	93.79
MSCN [32]	38.37M	**99.26±0.07**	**99.95±0.10**	93.28±0.26	98.21±0.39	92.76±0.13	93.67±0.29
STMSF [58]	48.85M	99.01±0.31	99.58±0.23	96.15±0.16	97.51±0.37	92.88±0.16	94.95±0.11
Ours	1.35M	98.89±0.16	99.81±0.10	**97.47±0.09**	**98.42±0.10**	**94.69±0.09**	**96.15±0.04**

experimental results indicate that the proposed method has strong feature representation capabilities under limited training samples. In addition, we only need to train MSA while keeping the VFM backbone frozen. The training parameters of VFM-FAA are only 1.35 M, which greatly reduces the training cost.

The advantages of VFM-FAA mainly come from the following two aspects. On the one hand, our proposed VFM-FAA employs the general visual feature representation of the VFM, enabling accurate remote sensing scene recognition. On the other hand, unlike existing deep learning methods that train deep networks from scratch, our proposal adapts the pretrained VFM for RSC by training a small portion of the model parameters, reducing the reliance on labeled training samples.

2) Comparison With PEFT: We further compare with representative PEFT methods on UCM, AID, and NWPU-RESISC45. As shown in Table II, our approach achieves the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:47:23 UTC from IEEE Xplore. Restrictions apply.

TABLE II
EXPERIMENTAL RESULTS OF DIFFERENT PEFT METHODS ON UCM, AID, AND NWPU-RESISC45 DATASETS

Methods	Trainable Params	UCM		AID		NWPU-RESISC45	
		50%	80%	20%	50%	10%	20%
Linear-probing	0.05M	94.86	95.95	92.45	94.98	86.42	88.71
Full-tuning	304.41M	98.86	99.52	97.71	98.08	94.61	95.91
Bia-tuning [59]	0.32M	97.05	98.81	95.69	97.62	91.26	93.62
LoRA [60]	2.01M	98.48	99.76	96.98	98.18	93.39	95.46
SSF [61]	0.14M	96.48	98.81	95.64	97.58	91.69	93.62
Conv-Adapter [62]	6.61M	93.62	96.19	89.18	94.56	83.75	88.97
AdaptFormer [63]	3.22M	97.90	99.52	97.15	98.14	93.82	95.50
VPT-Deep [64]	1.27M	98.19	99.05	95.89	97.64	91.53	94.00
Ours	1.35M	**98.89±0.16**	**99.81±0.10**	**97.47±0.09**	**98.42±0.10**	**94.69±0.09**	**96.15±0.04**

TABLE III
EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON MILLION-AID

Methods	10%	20%	50%	80%
EMTCAL [5]	90.46	93.76	95.62	97.45
MHCT [41]	89.68	92.73	95.42	96.70
MSCN [32]	87.33	92.28	95.30	97.45
Ours	**93.93**	**96.33**	**97.78**	**98.85**

TABLE IV
EXPERIMENTAL RESULTS OF DIFFERENT METHODS ACROSS DATASETS

Methods	NWPU → UCM	NWPU → AID	AID → UCM
EMTCAL [5]	69.16	77.20	59.14
MHCT [41]	70.53	75.11	59.71
MSCN [32]	67.68	75.14	57.14
Ours	**89.89**	**85.52**	**83.43**

best accuracy across all datasets. Compared with full fine-tuning, our method surpasses its performance while using far fewer trainable parameters, substantially reducing the training cost. In contrast, linear probing, although extremely parameter-efficient, falls behind by a clear margin, especially under limited supervision. Compared to existing PEFT methods, our method shows more consistent gains in low-data regimes, indicating that it adapts VFMs to remote sensing scenes more effectively.

D. Comparison on More Challenging Datasets

We conduct experiments on a larger dataset, Million-AID [65], to further validate the effectiveness of our method. As shown in Table III, our approach consistently outperforms prior methods across all training ratios. In particular, we achieve 93.93% accuracy using only 10% labeled data, surpassing the best competitor by 3.47%. The advantage remains significant at 20% and 50% supervision, with improvements of 2.57% and 2.1%, respectively. These results indicate that our method has stronger generalization on large and diverse remote sensing scenes.

E. Generalization Comparison Across Datasets

We further evaluate cross-dataset generalization by training on one dataset and testing on another. Table IV shows that our method consistently achieves the best performance under all transfer settings. Specifically, we obtain 89.89% on NWPU→UCM, 85.52% on NWPU→AID, and 83.43% on AID→UCM, outperforming the strongest baseline by 19.36%, 8.32%, and 23.72%, respectively. These results indicate that our approach learns more domain-invariant representations and generalizes better to unseen datasets with different scene distributions and imaging conditions.

F. Ablation Study

This part mainly verifies the effectiveness of each module in VFM-FAA, such as the MSA and MFA. The experimental results on three test datasets are shown in Table V.

1) Influence of the VFM: As shown in Table V, when we directly employ the pretrained VFM for RSC, the scene category of most remote sensing images can be correctly classified and identified. For example, the accuracies are 94.86% (50% training) and 95.24% (80% training) in the UCM dataset, are 92.44% (20% training) and 94.96% (50% training) in the AID dataset, and are 86.42% (10% training) and 88.70% (20% training) in the NWPU-RESISC45 dataset. These experimental results demonstrate that the pretrained VFM can extract general features from remote sensing images, effectively facilitating scene classification.

2) Influence of the MSA: First, the scene classification results on all test datasets are lowest without the MSA and MFA modules, as shown in Table V. The proposed MSA brings significant improvements in RSC, especially for complex remote sensing scenes. Specifically, in the UCM dataset, MSA boosts the scene classification accuracy by 3.61% (from 94.86% to 98.47%) based on 50% training samples. MSA increases the accuracy by 4.68% (from 92.44% to 97.12%) in the AID dataset based on 20% training samples. In the NWPU-RESISC45 dataset, MSA improves the classification accuracy

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:47:23 UTC from IEEE Xplore. Restrictions apply.

TABLE V
ABLATION STUDY ON THE MSA AND MULTILEVEL FEATURE AGGREGATION MODULE (MFA)

MSA	MFA	UCM		AID		NWPU-RESISC45	
		50%	80%	20%	50%	10%	20%
		94.86±0.00	95.24±0.00	92.44±0.00	94.96±0.00	86.42±0.00	88.70±0.00
✓		98.47±0.22	99.67±0.01	97.12±0.01	98.08±0.14	94.64±0.04	95.96±0.31
	✓	98.10±0.00	98.81±0.00	95.90±0.00	97.18±0.00	90.80±0.00	92.56±0.00
✓	✓	**98.89±0.16**	**99.81±0.10**	**97.47±0.09**	**98.42±0.10**	**94.69±0.09**	**96.15±0.04**

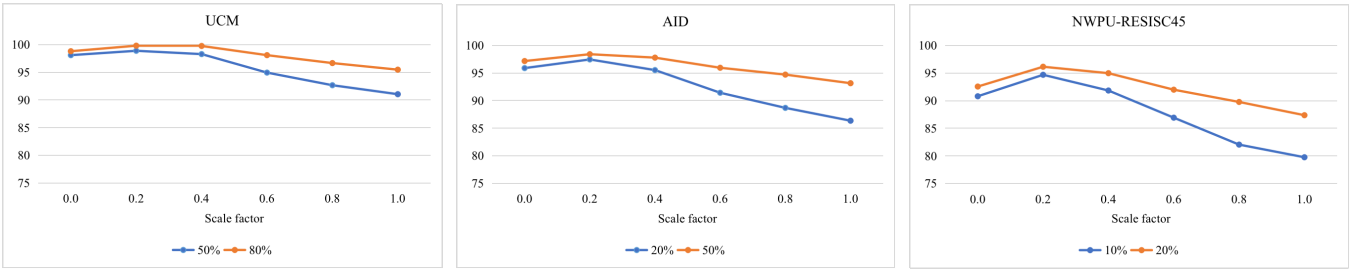


Fig. 7. Influence of the scale factor, s, in the MSA module on three test datasets. (a) OA% results on UCM. (b) OA% results on AID. (c) OA% results on NWPU-RESISC45.

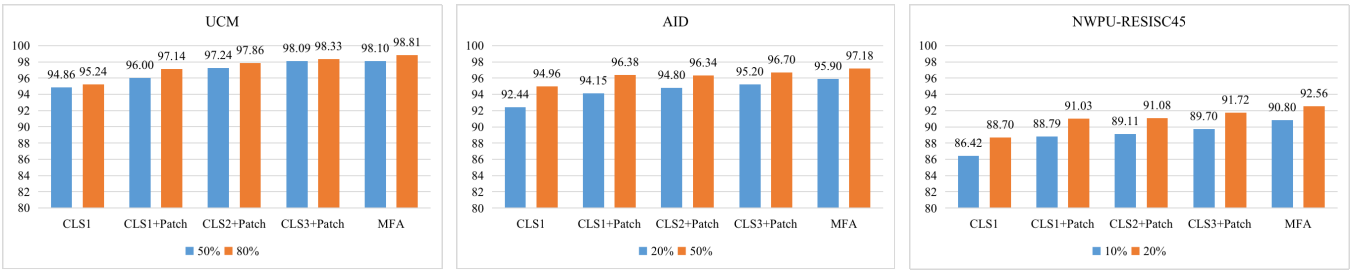


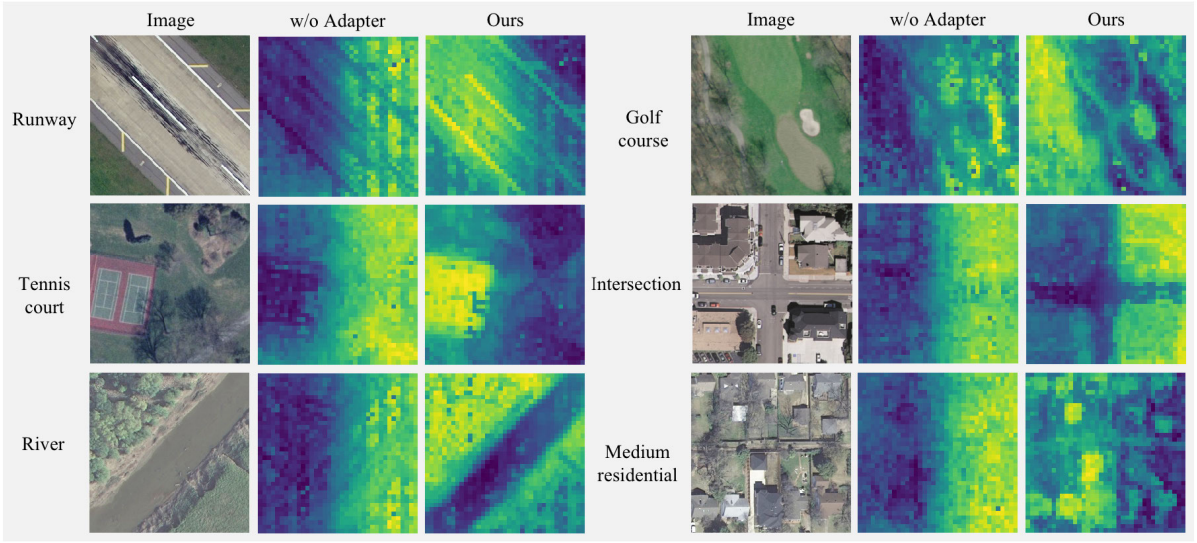
Fig. 8. Influence of the multilevel feature aggregation module on three test datasets. "CLS" means using the CLS token in the last level for scene category prediction, "CLS∗+Patch" means adopting the CLS token in the last ∗ levels and the patch token in the last level for scene classification, and "MFA" means fusing CLS tokens in the last four levels and the patch token in the last level for scene classification. (a) OA% results on UCM. (b) OA% results on AID. (c) OA% results on NWPU-RESISC45.

by 8.22% (from 86.42% to 94.64%) based on 10% training samples. MSA can effectively adapt the feature representation of the VFM for remote sensing images, significantly improving the scene classification performance. In addition, we present the influence of the scale factor s in the MSA module on three test datasets in Fig. 7. As the scale factor increases, the performance initially improves and then declines. When $s = 0.2$, the deep model achieves the best scene classification performance. Thus, we adopt $s = 0.2$ in experiments.

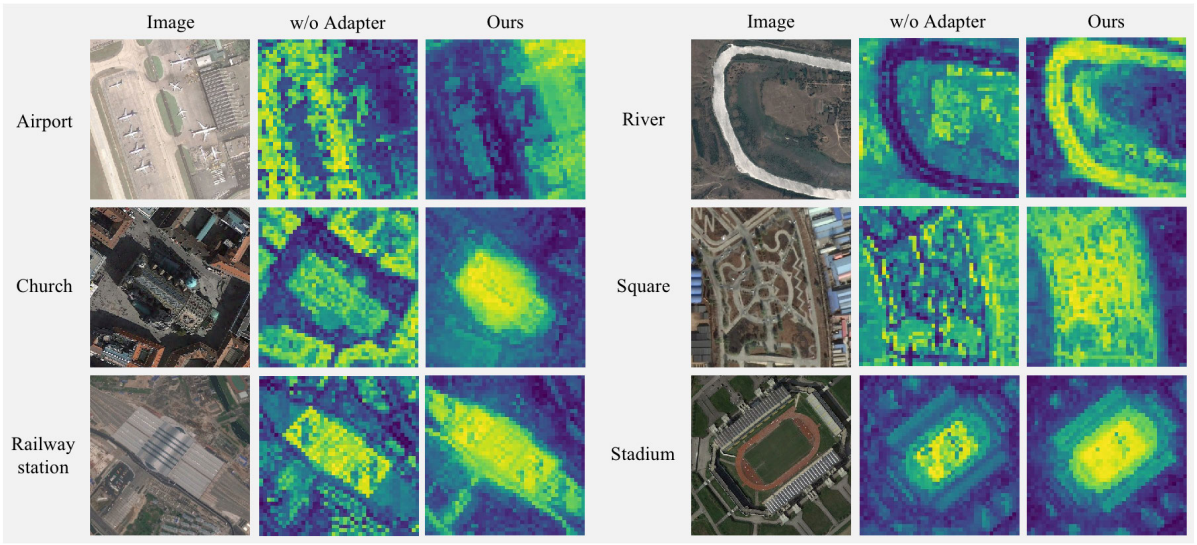
3) Influence of the MFA: From the results in Table V, we can see that the MFA can boost scene classification accuracy. For example, MFA improves the accuracy by 3.24% (from 94.86% to 98.10%) in the UCM dataset based on 50% training samples, increases the accuracy by 3.46% (from 92.44% to 95.90%) in the AID dataset based on 20% training samples, and improves the accuracy by 4.38% (from 86.42% to 90.80%) in the NWPU-RESISC45 dataset based on 10% training samples. MFA fuses multilevel semantic information, which can effectively enhance the perception ability for different remote sensing scenes.

We further explore the effectiveness of the multilevel CLS token and patch token in the MFA module. As shown in Fig. 8, when the deep model merely employs the CLS token in the last level for scene semantic category prediction, the accuracy on all test datasets is the smallest. When the deep model adopts the CLS token and patch token in the last level for scene classification, the performance is improved from 94.86% to 96.00% in the UCM dataset based on 50% training samples, increased from 92.44% to 94.15% in the AID dataset based on 20% training samples, and improved from 86.42% to 88.79% in the NWPU-RESISC45 dataset based on 10% training samples. These experimental results indicate the effectiveness of the spatial local information in the patch token for scene perception. Furthermore, the more levels of CLS tokens are fused in the MFA module, the better the scene classification performance. The introduced

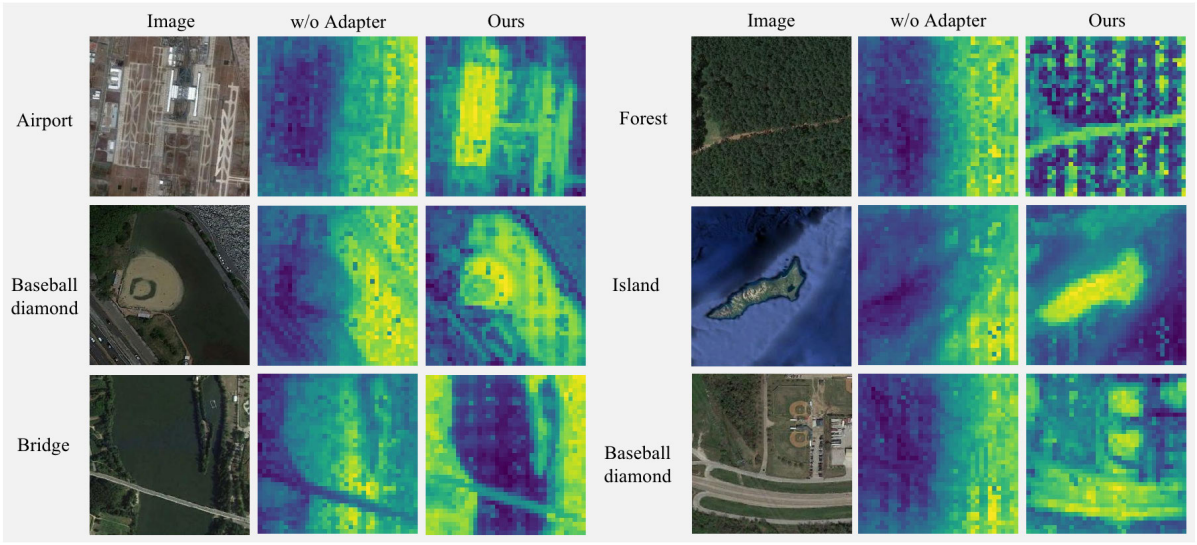
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:47:23 UTC from IEEE Xplore. Restrictions apply.



(a)



(b)



(c)

Fig. 9. Visualization results of the features learned by the VFM with the MSA module (Ours) and without the adapter (w/o adapter). (a) UCM. (b) AID. (c) NWPU-RESISC45.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:47:23 UTC from IEEE Xplore. Restrictions apply.

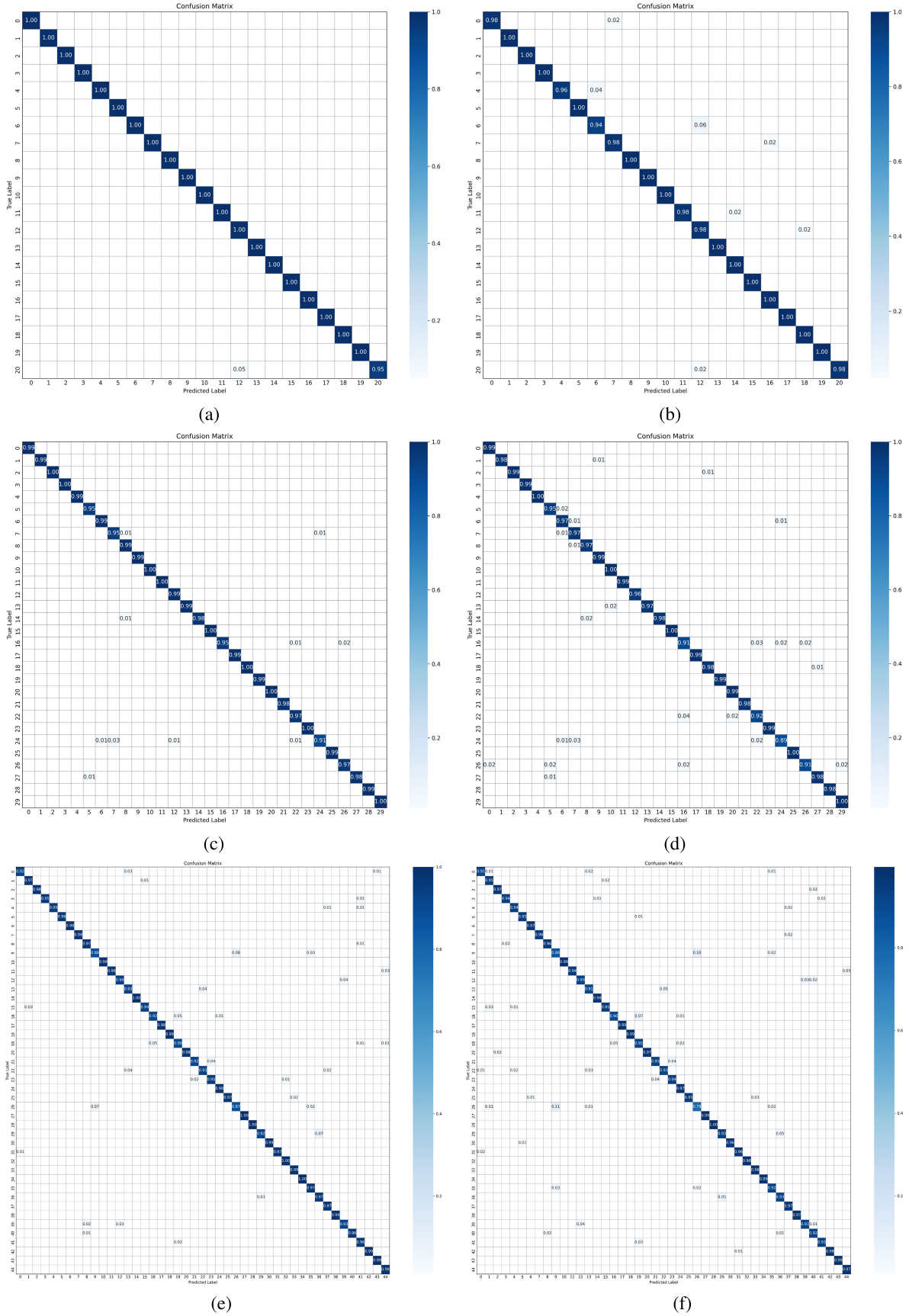


Fig. 10. Confusion matrices on three test datasets based on different ratios of training samples. (a) UCM 80% training samples. (b) UCM 50% training samples. (c) AID 50% training samples. (d) AID 20% training samples. (e) NWPU-RESISC45 20% training samples. (f) NWPU-RESISC45 10% training samples.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:47:23 UTC from IEEE Xplore. Restrictions apply.

multilevel CLS token can further significantly improve scene classification performances. For example, compared with the deep model with the CLS token in the last level, the deep model with the CLS token in the last four levels increases the accuracy from 96.00% to 98.10% in the UCM dataset based on 50% training samples, boosts the accuracy from 94.15% to 95.90% in the AID dataset based on 20% training samples, and improves the results from 88.79% to 90.80% in the NWPU-RESISC45 dataset based on 10% training samples.

Overall, the deep model with both MSA and MFA modules acquires the best performance. MSA mainly improves the scene classification performances from the perspective of feature adaptation, enabling the general feature representation to be more suitable for remote sensing images. MFA increases the scene classification accuracy from the perspective of scene semantic perception, capturing multilevel semantic information and boosting scene semantic category distinguishing. MSA brings larger accuracy improvements than MFA in RSC. This experimental result indicates that it is important to extract discriminative features suitable for remote sensing images and classification.

G. Visualization Results

1) Feature Visualizations: Fig. 9 shows the feature maps of different scenes extracted by the pretrained VFM and after feature adaptation. It can be seen that the pretrained VFM can capture the general visual features in most remote sensing images, such as edges and textures. However, the pretrained frozen VFM does not distinguish between the background and the object during the feature extraction process. Thus, there are a large number of background features, which will affect the discriminative features of objects related to the scene category, reducing the accuracy of scene classification.

After feature adaptation, our trained deep model pays more attention to the discriminative features of the objects related to the scene category. For example, as shown in Fig. 9(a), the deep model can capture the features of the way at the runway scene and the features of the tennis court. As shown in Fig. 9(b), our deep model can extract the features of the parallel tracks and platforms at the railway station scene, the architectural structure in the church scene, and the river morphology of the river scene. As shown in Fig. 9(c), our deep model with the feature adapter can accurately perceive the distinctive characteristics of the objects in various scenes, such as the airport, baseball, forest, and island. The MSA module enables the deep model to focus on discriminative features of objects relevant to the scene category rather than general feature representations, thereby improving the RSC accuracy.

2) CM Visualizations: Fig. 10 presents the confusion matrices of our proposal on UCM, AID, and NWPU-RESISC45 datasets, respectively. It can be seen that the deep model can accurately identify most remote sensing scenes. When there is a large number of training samples, the deep model can correctly identify each scene category. As the number of training samples decreases, there will be a very slight confusion between different scene categories. However, due to the introduction of a priori knowledge from the VFM, the performance of our proposal is not highly dependent on the optimization of the training samples, which can still achieve significant classification performance based on limited training samples. These visualization results demonstrate that our deep model based on VFM can extract discriminative features and has strong scene classification ability.

V. Conclusion

This article proposed a VFM-based framework with a multilevel feature adaptation and aggregation approach for RSC, VFM-FAA. We propose an MSA to improve the scene classification performances from the perspective of feature adaptation, which utilizes the general feature representation of the VFM and adapts the features to be more suitable for remote sensing images. In addition, this article builds a MFA to enhance the perspective of scene semantic perception, which captures multilevel semantic information for boosting scene semantic category distinguishing. Extensive experiments on three public datasets have demonstrated the feasibility of transferring pretrained VFM for RSC, the effectiveness of the proposed MSA and MFA, and the significant advantages of our proposal under limited training samples. In the future, we will study lightweight parameter-efficient fine-tuning methods for adapting the VFM for RSC.

References

[1] G. Cheng, X. Xie, J. Han, L. Guo, and G.-S. Xia, "Remote sensing image scene classification meets deep learning: Challenges, methods, benchmarks, and opportunities," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3735–3756, 2020.

[2] Y. Zhao, J. Liu, J. Yang, and Z. Wu, "EMSCNet: Efficient multisample contrastive network for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605814.

[3] C. Szegedy et al., "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1–9.

[4] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[5] X. Tang, M. Li, J. Ma, X. Zhang, F. Liu, and L. Jiao, "EMTCAL: Efficient multiscale transformer and cross-level attention learning for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5626915.

[6] G.-S. Xia et al., "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.

[7] J.-B. Grill et al., "Bootstrap your own latent-a new approach to self-supervised learning," in *Proc. 34th Int. Conf. Neural Inf. Process. Syst.*, 2020, pp. 21271–21284.

[8] A. Kirillov et al., "Segment anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 4015–4026.

[9] M. Oquab et al., "DINOv2: Learning robust visual features without supervision," 2023, *arXiv:2304.07193*.

[10] C. Huo et al., "When remote sensing meets foundation model: A survey and beyond," *Remote Sens.*, vol. 17, no. 2, p. 179, Jan. 2025.

[11] Z. Huang et al., "A survey on remote sensing foundation models: From vision to multimodality," 2025, *arXiv:2503.22081*.

[12] A. Xiao et al., "Foundation models for remote sensing and Earth observation: A survey," *IEEE Geosci. Remote Sens. Mag.*, vol. 13, no. 4, pp. 297–324, Dec. 2025.

[13] X. He, Y. Chen, L. Huang, D. Hong, and Q. Du, "Foundation model-based multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5502117.

[14] D. Wang et al., "Advancing plain vision transformer toward remote sensing foundation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5607315.

[15] J. Zhang, L. Ding, T. Zhou, J. Wang, P. M. Atkinson, and L. Bruzzone, "Recurrent semantic change detection in VHR remote sensing images using visual foundation models," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5402314.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:47:23 UTC from IEEE Xplore. Restrictions apply.

[16] K. Chen et al., "RSPrompter: Learning to prompt for remote sensing instance segmentation based on visual foundation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4701117.

[17] S. Lu et al., "Vision foundation models in remote sensing: A survey," *IEEE Geosci. Remote Sens. Mag.*, vol. 13, no. 3, pp. 190–215, Sep. 2025.

[18] P. Wang et al., "RingMoGPT: A unified remote sensing foundation model for vision, language, and grounded tasks," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5611320.

[19] E. Arnaudo et al., "FMARS: Annotating remote sensing images for disaster management using foundation models," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2024, pp. 3920–3924.

[20] M. Wang et al., "RSBuilding: Toward general remote sensing image building extraction and change detection with foundation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4707417.

[21] W. Liu, P. Duan, Z. Xie, X. Kang, and S. Li, "Learning from vision foundation models for cross-domain remote sensing image segmentation," *IEEE Trans. Image Process.*, vol. 34, pp. 4553–4565, 2025.

[22] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," 2023, *arXiv:2311.07113*.

[23] X. Guo et al., "SkySense: A multi-modal remote sensing foundation model towards universal interpretation for Earth observation imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 27662–27673.

[24] X. Sun et al., "RingMo: A remote sensing foundation model with masked image modeling," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2022, Art. no. 5612822.

[25] J. Guo et al., "C^3DA: A universal domain adaptation method for scene classification from remote sensing imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[26] X. Tang, Q. Ma, X. Zhang, F. Liu, J. Ma, and L. Jiao, "Attention consistent network for remote sensing scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2030–2045, 2021.

[27] W. Wang et al., "PVT v2: Improved baselines with pyramid vision transformer," *Comput. Vis. Media*, vol. 8, no. 3, pp. 415–424, Sep. 2022.

[28] J. Ma, M. Li, X. Tang, X. Zhang, F. Liu, and L. Jiao, "Homo–heterogenous transformer learning framework for RS scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 2223–2239, 2022.

[29] Q. Wang, W. Huang, Z. Xiong, and X. Li, "Looking closer at the scene: Multiscale representation learning for remote sensing image scene classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 4, pp. 1414–1428, Apr. 2022.

[30] S. Chavda and M. Goyani, "Scene level image classification: A literature review," *Neural Process. Lett.*, vol. 55, no. 3, pp. 2471–2520, Jun. 2023.

[31] R. Du, X. Tang, J. Ma, X. Zhang, F. Liu, and L. Jiao, "Semantic-assisted feature integration network for multilabel remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5603015.

[32] J. Ma, W. Jiang, X. Tang, X. Zhang, F. Liu, and L. Jiao, "Multiscale sparse cross-attention network for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5605416.

[33] S. A. B. Pacheco, M. Goyani, Z. G. Rehman, S. F. Rehman, T. Champaneria, and S. Goyani, "Enhanced content-based image retrieval using multivisual features fusion," *Int. J. Comput. Appl.*, vol. 47, no. 10, pp. 835–856, Oct. 2025.

[34] Y. Yang, X. Tang, Y.-M. Cheung, X. Zhang, and L. Jiao, "SAGN: Semantic-aware graph network for remote sensing scene classification," *IEEE Trans. Image Process.*, vol. 32, pp. 1011–1025, 2023.

[35] X. Zhang, X. Fan, G. Wang, P. Chen, X. Tang, and L. Jiao, "MFGNet: Multibranch feature generation networks for few-shot remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5609613.

[36] Y. Yu, S. Wang, and Z. Fu, "Prompt-integrated adversarial unsupervised domain adaptation for scene recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4402312.

[37] Y. Zhao, J. Liang, S. Huang, and P. Huang, "Hierarchical deep features progressive aggregation for remote sensing images scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 9442–9450, 2024.

[38] X. Wang, S. Wang, C. Ning, and H. Zhou, "Enhanced feature pyramid network with deep semantic embedding for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7918–7932, Sep. 2021.

[39] G. Zhang et al., "A multiscale attention network for remote sensing scene images classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 9530–9545, 2021.

[40] S. Chavda and M. Goyani, "Multi-label remote sensing scene classification using two-level double channel spatial attention residual blocks," *J. Appl. Remote Sens.*, vol. 18, no. 3, Sep. 2024, Art. no. 036511.

[41] D. Zhang, W. Ma, L. Jiao, X. Liu, Y. Yang, and F. Liu, "Multiple hierarchical cross-scale transformer for remote sensing scene classification," *Remote Sens.*, vol. 17, no. 1, p. 42, Dec. 2024.

[42] X. Tang, W. Lin, J. Ma, X. Zhang, F. Liu, and L. Jiao, "Class-level prototype guided multiscale feature learning for remote sensing scene classification with limited labels," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5622315.

[43] R. Du, X. Tang, J. Ma, X. Zhang, and L. Jiao, "MLMamba: A mamba-based efficient network for multi-label remote sensing scene classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 7, pp. 6245–6258, Jul. 2025.

[44] A. Ma, N. Yu, Z. Zheng, Y. Zhong, and L. Zhang, "A supervised progressive growing generative adversarial network for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5618818.

[45] Z. Ji, L. Hou, X. Wang, G. Wang, and Y. Pang, "Dual contrastive network for few-shot remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605312.

[46] F. Tian et al., "HiReNet: hierarchical-relation network for few-shot remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5603710.

[47] Y. Li, Z. Zhu, J.-G. Yu, and Y. Zhang, "Learning deep cross-modal embedding networks for zero-shot remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10590–10603, Dec. 2021.

[48] C. Liu, S. Ma, Z. Li, W. Yang, and Z. Han, "Mining contrastive relations between cross-modal features for zero-shot remote sensing image scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[49] B. Chai, T. Zhao, R. Yang, N. Zhang, T. Tian, and J. Tian, "Label semantic dynamic guidance network for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5610414.

[50] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "SimCLR: A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn. (PMLR)*, New York, NY, USA, 2020, vol. 2, no. 4, pp. 1597–1607.

[51] Z. Xie et al., "SimMIM: A simple framework for masked image modeling," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 9653–9663.

[52] C. Pan et al., "Vision foundation model guided multimodal fusion network for remote sensing semantic segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 9409–9431, 2025.

[53] X. Zou et al., "Adapting vision foundation models for robust cloud segmentation in remote sensing images," 2024, *arXiv:2411.13127*.

[54] N. Houlsby et al., "Parameter-efficient transfer learning for NLP," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 2790–2799.

[55] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. 9th Int. Conf. Learn. Represent.*, Oct. 2020, pp. 1–22.

[56] L. Yuan et al., "Tokens-to-token ViT: Training vision transformers from scratch on ImageNet," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2021, pp. 558–567.

[57] Z. Dong, Y. Gu, and T. Liu, "UPetu: A unified parameter-efficient fine-tuning framework for remote sensing foundation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5616613.

[58] Y. Duan, C. Song, Y. Zhang, P. Cheng, and S. Mei, "STMSF: Swin transformer with multi-scale fusion for remote sensing scene classification," *Remote Sens.*, vol. 17, no. 4, p. 668, Feb. 2025.

[59] E. Ben Zaken, Y. Goldberg, and S. Ravfogel, "BitFit: Simple parameter-efficient fine-tuning for transformer-based masked language-models," in *Proc. 60th Annu. Meeting Assoc. Comput. Linguistics*, vol. 2, 2022, pp. 1–9.

[60] E. J. Hu et al., "LoRA: Low-rank adaptation of large language models," in *Proc. ICLR*, 2022, vol. 1, no. 2, p. 3.

[61] J. Feng, D. Lian, X. Wang, and D. Zhou, "Scaling & shifting your features: A new baseline for efficient model tuning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 109–123.

[62] H. Chen et al., "Conv-adapter: Exploring parameter efficient transfer learning for ConvNets," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2024, pp. 1551–1561.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:47:23 UTC from IEEE Xplore. Restrictions apply.

[63] S. Chen et al., "AdaptFormer: Adapting vision transformers for scalable visual recognition," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 16664–16678.
[64] M. Jia et al., "Visual prompt tuning," in *Proc. Eur. Conf. Comput. Vis.*, Oct. 2022, pp. 709–727.
[65] Y. Long et al., "On creating benchmark dataset for aerial image interpretation: Reviews, guidances, and million-AID," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4205–4230, 2021.



Chonghua Lv (Student Member, IEEE) received the B.S. degree from Xi'an University of Posts and Telecommunications, Xi'an, China, in 2021. He is currently pursuing the Ph.D. degree with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education of China, Xidian University, Xi'an.

His research interests include deep learning, knowledge distillation, and semantic segmentation.



Dan Zhang received the B.S. degree in intelligent science and technology, the M.S. degree in circuits and systems, and the Ph.D. degree in computer science and technology from Xidian University, Xi'an, China, in 2011, 2016, and 2025, respectively.

Since 2016, she has been the Deputy Director of the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education of China. Her research interests include image processing and machine learning.



Dou Quan (Member, IEEE) received the B.S. and Ph.D. degrees from Xidian University, Xi'an, China, in 2015 and 2021, respectively.

She is currently an Associate Professor with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education of China, School of Artificial Intelligence, Xidian University. Her research interests include machine learning, deep learning, metric learning, image matching, image registration, and image classification.



Ning Huyan (Member, IEEE) received the B.S. and Ph.D. degrees from Xidian University, Xi'an, China, in 2015 and 2022, respectively.

From 2019 to 2020, he was a Joint Ph.D. Student under the supervision of Prof. Jocelyn Chanussot at the Inria Grenoble Rhône-Alpes Research Center, Montbonnot-Saint-Martin, France. From 2022 to 2024, he was a Researcher at Sensetime, Xi'an. He is currently a Post-Doctoral Researcher with the Department of Automation, Tsinghua University, Beijing, China. His research interests include hyperspectral image anomaly detection, human pose estimation, and motion generation.



Jiaqi Chen is pursuing the B.S. degree in with the School of Artificial Intelligence, Xidian University, Xi'an, China.

He has been admitted to pursue a master's degree at the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education, School of Artificial Intelligence, Xidian University, Xi'an, China. His research interests include knowledge distillation and deep learning for remote sensing.



Shuang Wang (Senior Member, IEEE) received the B.S., M.S., and Ph.D. degrees in circuits and systems from Xidian University, Xi'an, China, in 2000, 2003, and 2007, respectively.

She is currently a Professor with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education of China, Xidian University. Her research interests include sparse representation, image processing, synthetic aperture radar (SAR) automatic target recognition, remote sensing image captioning, and polarimetric SAR data analysis and interpretation.



Licheng Jiao (Life Fellow, IEEE) was born in Shaanxi, China, in October 1959. He received the B.S. degree from Shanghai Jiao Tong University, Shanghai, China, in 1982, and the M.S. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 1984 and 1990, respectively.

From 1984 to 1986, he was an Assistant Professor with the Civil Aviation Institute of China, Tianjin, China. From 1990 to 1991, he was a Post-Doctoral Fellow with the Key Laboratory for Radar Signal Processing, Xidian University, Xi'an. He is currently the Director of the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education of China. His research interests include signal and image processing, nonlinear circuits and systems theory, learning theory and algorithms, optimization problems, wavelet theory, and machine learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:47:23 UTC from IEEE Xplore. Restrictions apply.